Count Unique Characters of All Substrings of a Given String (hard)

Problem Statement

Given a string `s`, return the sum of `unique` characters in all possible substrings of a given string. You also need to count the unique characters of `repeating` substrings.

Examples

- **Example 1**:
  - **Input**: "aac"
  - **Expected Output**: 8
  - **Justification**: The substrings are "a", "a", "c", "aa", "ac", "aac". Their unique characters count are 1, 1, 1, 0, 2, 1 respectively. Summing up the unique characters: 1 + 1 + 1 + 0 (from "aa") + 2 (from "ac") + 1 (from "aac") = 6.
- **Example 2**:
  - **Input**: "aaa"
  - **Expected Output**: 6
  - **Justification**: The substrings are "a", "a", "a", "aa", "aa", and "aaa". Their unique characters count are 1, 1, 1, 0, 0, 0 respectively. So, there are total `3` unique characters in all substrings.
- **Example 3**:
  - **Input**: "abca"
  - **Expected Output**: 18
  - **Justification**: The substrings and their unique character counts are as follows: "a" (1), "b" (1), "c" (1), "a" (1), "ab" (2), "bc" (2), "ca" (2), "abc" (3), "bca" (3), "abca" (2). The sum of unique characters is **18**.

Try it yourself

Try solving this question here:

Java

```java
import java.util.Arrays;

public class Solution {
    public int uniqueLetterString(String S) {
        // ToDo: Write Your Code Here.
        return 0;
    }
}
```

Test Case | Run Code Result | Stdout

"AAC"

Use Example Testcases | Run | Submit

← Previous

Solution: Sliding Window Maximum

Next →

Solution: Count Unique Characters Of All Substrings Of A Given String

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